SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	**Total Marks:**	100
CO2 Statement:	*Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)*		
Student Name:	D. Hemasree	**Reg. No.:**	192524114
Section / Batch:	2025	**Date:**	

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Code of Conduct

I D. Hemasree/192524114 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: D. Hemy

Q1) Binary Search - Finding 34 in a Sorted Dataset.

Problem Interpretation:

Given a sorted array: [4, 9, 14, 24, 29, 34, 39, 44] (9 elements, index 0 to 8).

Goal: locate the target value 34 using binary search.

Since the array is already sorted, we can repeatedly eliminate half the search space instead of scanning one by one.

Why Binary Search Fits here:

Binary search works only on sorted data - this dataset satisfies that condition.

At each step, comparing the middle element with the target tells us which half to discard, cutting the problem size in half every time.

This makes it far more efficient than linear search for large datasets.

S.No	low	high	mid = (low + high) / 2	arr (mid)	comparision result.
1	0	8	4	24	24<34 → Search right half, low = 5
2	5	8	6	34	34 = 34 → Element found at Index 6.

Total comparision made = 2

Time complexity Analysis

- Each comparision halves the search interval
- Generate Formula: comparisions ≈ $\log_2(n)$ where $n = 9$

→ $\log_2(9) \approx 3.17$, so max 4 comparisions in worst case.

Time complexity: $O(\log n)$

Space complexity: $O(1)$ (iterative version, no extra memory used).

Verification & Efficiency:

arr[6] = 34, which matches the target exactly → result varified as correct.

only 2 comparisons were needed out of 9 elements, instead of checking out to 9 elements one by one (Linear search).

This confirms Binary search efficiency: It drastically reduces the number of checks needed for sorted data, especially useful as data set size grows.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________